Akshay Dhar

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AI Product Manager who takes AI from problem to production and gates every release on an evaluation. At Catalysk I owned a transaction categorization engine that auto-categorizes when confident and routes uncertain rows to human-in-the-loop review, cutting uncategorized transactions from 39% to 13%. At 9fin I shipped the Gemini API into production automation with self-written prompts, and owned an ML crawler delivering £390K in annual savings. I have since built five AI products with Claude Code, each with a PRD and held-out evals.

Core Competencies

- **AI Product Ownership**: an AI categorization engine taken from problem diagnosis to production; PRDs with north-star and guardrail metrics and release gates; precision trade-off calls backed by evidence.
- **Evaluations & Release Bars**: golden sets, held-out sets frozen before tuning, benchmarks, ablations against rules-only baselines and confidence gating; embedding model bake-offs (MiniLM, E5, BGE).
- **LLMs, AI Agents & RAG**: Gemini API in production with self-written prompts; in personal projects, an LLM agent with MCP tool calling and RAG on pgvector, LLM answers verified against source documents, Hindi and Bengali classification, and prompt-injection guards.
- **Building**: Claude Code and Cursor; Python (Pandas), SQL/PostgreSQL, AWS, DuckDB, FastAPI, n8n, GitHub Actions, REST APIs, config-driven YAML pipelines, 90+ automated data quality checks.
- **Domains**: fintech (UPI/VPA, P2P, institutional credit), ESG and sustainability, energy, edtech; in personal projects, RBI regulation, health insurance and product analytics.

Professional Experience

- **Catalysk, AI Sustainability Analytics** — **Bengaluru, India**
  Data Consultant — *Apr 2026 - Sep 2026*
  - Architected and led the build of a multi-stage categorization engine with a small team (rules, merchant extraction, an SBERT fallback and the evaluation harness), owning it from design through production.
  - Cut uncategorized transactions from 39% to 13% by redesigning UPI (VPA) and P2P categorization logic feeding client ESG reporting.
  - Gated auto-categorization at ~84% model confidence, routing uncertain rows to analyst review (human-in-the-loop) to prioritise precision over vanity coverage.
  - Defined the release bar with a 388-row UAT golden set and a 1,400-row benchmark across 7 bank statements, gating every model and taxonomy change.
  - Lifted category accuracy 5 points with an embedding model bake-off (MiniLM, E5, BGE) against the golden set, rejecting a slower model that only looked better on paper.
  - Grew model coverage from 21 to 36 categories and 10K to 17K training examples by retraining SBERT and merging a second tenant's taxonomy, with zero golden-set regression.
  - Cut new banking partner onboarding from days to hours with a config-driven YAML framework and one ingestion contract for CSV, Excel, JSON and REST, so analysts change rules without engineering.
  - Recovered a catch-all bucket at ~30% of volume by diagnosing a silent regression that disabled SBERT and 4 downstream stages; cut processing from hours to minutes per 1,000 rows with embedding caching.
  - Shipped 90+ automated data quality checks across 5 financial schemas, and an 18-table PostgreSQL observability layer with live Metabase dashboards replacing manual Excel review.

- **9fin, Institutional Financial Technology Platform** — **Belfast, UK**
  Senior Data Associate — *Jan 2025 - Sep 2025*
  - Delivered £390K annual cost savings and 32% faster client delivery by owning a production ML market-news crawler, resolving incidents and adding verification controls.
  - Accelerated client onboarding by 15% by shipping the Gemini API into production automation for post-acquisition schema alignment, with self-written prompts.
  - Mentored 15 analysts on data quality standards, verification workflows and tooling.
- **9fin, Institutional Financial Technology Platform** — **Belfast, UK**
  Data Associate — *May 2024 - Dec 2024*
  - Maintained 100% data accuracy against delivery SLAs across 140+ institutional credit instruments.
  - Built real-time KPI and incident dashboards (Power BI, Notion) for technical and non-technical stakeholders.
- **JamBird Marketing** — **Belfast, UK**
  Data Science Intern — *Jun 2023 - Sep 2023*
  - Deployed an NLP text classification model at 87% accuracy, replacing a manual review process.
  - Engineered features across 3M+ data points and presented findings to senior stakeholders.
- **BYJU'S** — **India**
  Data Analyst — *Aug 2020 - Jul 2021*
  - Implemented ML learning-path recommendations, improving student segmentation accuracy by 15%.
  - Optimised PySpark ETL pipelines, improving data processing time by 20%.

Personal Projects

- **Why Did It Drop?** *github.com/AKSHAYKUMARDHAR/Why-Did-It-Drop* — *Oct 2026*
  Agent that explains why a product metric moved, on Google's GA4 data (4.3M events, DuckDB). Failed its gate on a 70-case benchmark (20% wrong causes vs 5%); the model added no decision value over rules.
- **RBI Circular Radar**, *github.com/AKSHAYKUMARDHAR/RBI-Circular-Radar* — *Oct 2026*
  Live digest of which new RBI notifications apply to 17 entity types, what to do and by when, quoted from RBI's PDF. Code checks cut wrong dates from 11 to 1 of 165; blocked twice by its own zero-miss gate.
- **Will My Policy Pay?** *github.com/AKSHAYKUMARDHAR/Will-My-Policy-Pay* — *Oct 2026*
  Indian health insurance fine print, each limit quoted from the insurer's PDF and checked in code. Checks cut wrong facts from 8.2% to 1.2% (dev); 91.4% correct on 152 held-out facts, but blocked twice by its 2% gate.
- **Is This a Scam?** *github.com/AKSHAYKUMARDHAR/Is-This-A-Scam* — *Oct 2026*
  Live scam checker in Hindi, Bengali and English that says "can't tell" instead of guessing. Hand-written held-out sets: 1 false alarm in 114 genuine messages, 0 of 96 scams cleared; gate failed twice.
- **UPI Transaction Triage Agent**, *github.com/AKSHAYKUMARDHAR/UPI-Triage-Agent* — *Oct 2026*
  LLM agent with MCP tool calling, RAG on pgvector, a human review queue and an injection guard. On 57 unseen rows: 98.0% precision at 86% automation; 0% of unknown payees posted wrong (44.4% without RAG).

Education

- **Queen's University Belfast** — **Belfast, UK**
  MSc Data Analytics — *2022-2023*
- **Assam University Silchar** — **Silchar, India**
  BSc Computer Science — *2017-2020*

Certifications, Languages & Awards

- Oracle Cloud Infrastructure AI Foundations Associate; Google Data Analytics Professional Certificate; BCG GenAI and Data Science Job Simulations (Forage); Best Project, Irish Intervarsity (2023).
- English (full professional), Hindi (native), Assamese (native), Bengali (native).

References

- Available upon request.